deviations possible. But this time, we told the team to go for 6-sigma deviations, leaving potential for absolute failures of systems and engines, overcome them in flight and still land.

Q. You looked very confident in the final stages of the flight. Where did that sense of calmness come from?

I am not in the habit of getting excited. I have seen over 50 rocket launches in my career, with several ups and downs. My approach is that we have already done whatever we could do and can't do anything more. So, we must be ready for any kind of result and be able to handle failure if it happens.

Q. After Chandrayaan, what?

A lot of discussion is going on internally on what we can do further. Any mission to the moon should be based on the scientific goals that we set. Once all the Chandrayaan-3 data is in and analysed, there will be certain new questions that will need to be answered and we will have to derive new ways of doing them. Bring back some lunar soil, for instance, to study it better. Another thing under discussion is launching a large number of small satellites to study the moon. We are also involved in a joint programme with Japan's space agency, where we provide the lander and the Japanese the rover.

Q. How will the US-backed Artemis Accord, to which India too is a signatory now, help?

Whatever we have signed is part of the accord, not the Artemis Programme. The Artemis Programme is a project started by the US government for the return of human space flight missions to the moon to create a lunar gateway. As part of the accord, many nations, including India, have signed up with the US on a principle that is non-binding on either party. Both have agreed to follow certain rules and regulations, especially for sustainability, while exploring the moon and other bodies. Currently, being part of the Artemis Accord gives India certain privileges, including access to US technology, so that we can build more complex systems. It could be a big game-changer as far as technology transfer from the US to India is concerned, including for our private sector.

Q. There is renewed interest in the moon. Apart from America, the Chinese and the Russians as well as a host of other countries, are preparing to launch missions. What explains it?

> "Ultimately, our aim is to recover rockets, as SpaceX does, and also meet the future demand for heavy-lift vehicles"

Earlier, going to the moon was all about geopolitical competition and projection power and prowess. The cost of going to the moon too was extremely high. Today, it is not. The technology now is much more advanced; we can do robotic experiments better than what Apollo or the Soviets could. The other aspect is that human beings want to explore the universe, and the moon is a good staging ground for it. Interplanetary explorations, such as to Mars and Venus, need a reasonably long-term base on the moon.

Q. The Russians have rockets that can send a lander to the Moon within days rather than weeks as we do. As do the Americans. What do we need to do to upgrade our capabilities?

We need to have new propulsion technologies of course. Currently, we depend on solid, liquid and cryogenic motors, but if we have to really make it big in terms of launch capability, then we have to have high-pressure liquid engines that can lift a 50-tonne payload instead of the current 6-10 tonnage. We need to build one such engine and then make a cluster of them to provide the necessary thrust to carry the higher tonnage. Ultimately, our goal is to recover rockets, like SpaceX does, and also meet the future demand for heavy-lift vehicles.

Q. The Modi government has opened up space for greater private participation, including to build launchers and satellites. What role will ISRO play?

When you look at NASA, it already had a solid defence industrial sector working to develop space systems in the US. But, in India, ISRO is doing all the space work, including design capability and getting the manufacturing done in both public and private sector industry. With the Union government opening up space, we want to make space a commercially attractive venture. Unless it is profitable, private companies won't take the risk. So, we have to look at the many ways to do so, but ISRO must remain a technology developer of knowledge, supporter of this knowledge and an enabler for the private sector. Big players in India, who have enough resources and wealth, like L&T, the Tatas and Adanis, should get into the space business. We must also encourage young entrepreneurs to build startups for space. Thirdly, apart from rocket- and satellite-building, there is a lot of money in space application and downstream work. If you look at the space economy, then commercial rocket-launching is 15 per cent of the business, and satellite-building around 15-20 per cent. So, almost 60-70 per cent of new business is being generated in space applications and downstream activity. This is a slow transition, and ISRO has to support it. ■